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Transferring fuel between fuel reservoirs of an aircraft fuel system

Abstract

A method of operation is provided during which fuel is delivered to a powerplant using a fuel system. The powerplant and the fuel system are arranged onboard an aircraft. The fuel system includes a first fuel reservoir and a second fuel reservoir. The first fuel reservoir contains a quantity of the fuel. At least some of the quantity of the fuel contained in the first fuel reservoir is transferred into the second fuel reservoir when a fault is identified associated with the first fuel reservoir.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3982399	12/1975	Rookey	N/A	N/A
5660358	12/1996	Grafwallner	N/A	N/A
11063442	12/2020	Crawford	N/A	N/A
11258081	12/2021	Min	N/A	N/A
11738638	12/2022	Koo	N/A	N/A
11780597	12/2022	Swann	244/135R	B64D 27/10
11796132	12/2022	Saha	N/A	N/A
11828417	12/2022	Clarke	N/A	N/A
12031493	12/2023	Swann	N/A	F02C 3/24
2022/0316410	12/2021	Swann	N/A	F02C 9/40
2022/0324583	12/2021	Miftakhov	N/A	N/A
2023/0015930	12/2022	Meshkin Fam	N/A	B64D 37/30
2023/0193835	12/2022	Swann	N/A	B64D 27/10
2023/0323820	12/2022	Swann	60/39.094	F02C 9/26
2023/0339621	12/2022	Sarkar	N/A	F02C 3/22
2023/0356856	12/2022	Gilmore	N/A	N/A
2025/0153857	12/2024	Porterfield	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2023074866	12/2022	WO	N/A
2025114686	12/2024	WO	N/A

OTHER PUBLICATIONS

EP search report for EP25154261.9 dated Jun. 24, 2025. cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates generally to an aircraft and, more particularly, to a fuel system for the aircraft.

BACKGROUND INFORMATION

(2) An aircraft may include a fuel system for delivering fuel to a powerplant such as a gas turbine engine for an aircraft propulsion system. Various types and configurations of fuel systems are

known in the art. While these known fuel systems have various benefits, there is still room in the art for improvement.

SUMMARY

(3) According to an aspect of the present disclosure, a method of operation is provided during which fuel is delivered to a powerplant using a fuel system. The powerplant and the fuel system are arranged onboard an aircraft. The fuel system includes a first fuel reservoir and a second fuel reservoir. The first fuel reservoir contains a quantity of the fuel. At least some of the quantity of the fuel contained in the first fuel reservoir is transferred into the second fuel reservoir when a fault is identified associated with the first fuel reservoir.

(4) According to another aspect of the present disclosure, an assembly is provided for an aircraft. This assembly includes an aircraft powerplant and a fuel system. The fuel system includes a supply circuit, an array of fuel reservoirs and a fuel control system. The array of fuel reservoirs includes a first fuel reservoir and a second fuel reservoir. The first fuel reservoir and the second fuel reservoir are fluidly couplable to the supply circuit in parallel. The fuel control system is configured to direct fuel stored in the array of fuel reservoirs to the aircraft powerplant through the supply circuit. The fuel control system is configured to transfer at least some of a quantity of the fuel contained in the first fuel reservoir into the second fuel reservoir following identification of a fault associated with the first fuel reservoir while the fuel continues to be directed to the aircraft powerplant through the supply circuit.

(5) According to still another aspect of the present disclosure, another assembly is provided for an aircraft. This assembly includes an aircraft powerplant and a fuel system. The fuel system includes a supply circuit, an array of fuel reservoirs, a fuel delivery system and a fuel transfer system discrete from the fuel delivery system. The array of fuel reservoirs includes a first fuel reservoir and a second fuel reservoir. The first fuel reservoir and the second fuel reservoir are fluidly couplable to the supply circuit in parallel. The fuel delivery system is configured to direct hydrogen fuel stored in the array of fuel reservoirs to the aircraft powerplant through the supply circuit. The fuel transfer system is configured to transfer at least some of a quantity of the hydrogen fuel contained in the first fuel reservoir into the second fuel reservoir in response to identification of a fault associated with the first fuel reservoir.

(6) The fuel system may be a hydrogen fuel system. The first fuel reservoir may be a first liquid hydrogen reservoir. The second fuel reservoir may be a second liquid hydrogen reservoir.

(7) The fuel may be or otherwise include hydrogen fuel.

(8) At least some (or all) of the fuel transferred from the first fuel reservoir into the second fuel reservoir may be in a gaseous phase.

(9) At least some (or all) of the fuel transferred from the first fuel reservoir into the second fuel reservoir may be in a liquid phase.

(10) The method may also include a step of identifying the fault associated with the fuel first reservoir.

(11) The fault may be a fuel leak.

(12) The fault may be indicative of a fuel leak from the first fuel reservoir.

(13) The fault may be indicative of a fuel leak from a fuel line fluidly coupled to the first fuel reservoir.

(14) The method may also include a step of fluidly decoupling the first fuel reservoir from a supply circuit of the fuel system when the fault is identified associated with the first fuel reservoir. The delivering of the fuel may include delivering the fuel to the powerplant through the supply circuit.

(15) The fuel may be delivered to the powerplant using the fuel system before, during and after the transferring of the fuel from the first fuel reservoir into the second fuel reservoir.

(16) The transferring may include pumping at least some (or all) of the quantity of the fuel contained in the first fuel reservoir into the second fuel reservoir.

(17) The transferring may include opening a valve between the first fuel reservoir and the second

fuel reservoir to equalize a fuel parameter between the first fuel reservoir and the second fuel reservoir.

(18) The transferring of the fuel from the first fuel reservoir into the second fuel reservoir may increase a quantity of the fuel contained in the second fuel reservoir above a predetermined maximum capacity of the second fuel reservoir.

(19) The method may also include a step of transferring some (or all) of the quantity of the fuel contained in the first fuel reservoir into a third fuel reservoir of the fuel system when the fault is identified associated with the first fuel reservoir.

(20) The method may also include a step of venting some (or all) of the quantity of the fuel contained in the first fuel reservoir to an environment external to the aircraft during and/or following the transferring of the fuel from the first fuel reservoir into the second fuel reservoir.

(21) The powerplant may be configured as or otherwise include a heat engine.

(22) The powerplant may be configured as or otherwise include a fuel cell.

(23) The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.

(24) The foregoing features and the operation of the invention will become more apparent in light of the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic illustration of a system onboard an aircraft with a fuel system servicing a powerplant.

(2) FIG. 2 is a flow diagram of a method of operation for an aircraft system.

DETAILED DESCRIPTION

(3) FIG. 1 illustrates a system **10** for an aircraft **12**. The aircraft **12** may be an airplane, a helicopter, a drone (e.g., an unmanned aerial vehicle (UAV)) or any other manned or unmanned aerial vehicle or system. The aircraft system **10** is arranged onboard the aircraft **12** and is configured of use while the aircraft **12** is in flight and/or on ground. The aircraft system **10** of FIG. 1 includes a powerplant **14** for the aircraft **12** and a (e.g., a hydrogen) fuel system **16** for the aircraft **12** and its powerplant **14**. The aircraft system **10** of FIG. 1 also includes a sensor system **18** and a controller **20**.

(4) The aircraft powerplant **14** may be configured as, or otherwise included as part of, a propulsion system for the aircraft **12**. The aircraft powerplant **14** may also or alternatively be configured as, or otherwise included as part of, an electrical power system for the aircraft **12**. The aircraft powerplant **14** may include a heat engine such as, but not limited to, a gas turbine engine, a rotary engine (e.g., a Wankel engine), a reciprocating piston engine or any other continuous or intermittent internal combustion (IC) engine. Examples of the gas turbine engine include a turboprop engine, a turbojet engine, a turboprop engine and a turboshaft engine. The aircraft powerplant **14** may also or alternatively include a fuel cell. The present disclosure, however, is not limited to the foregoing exemplary powerplant types or configurations.

(5) The fuel system **16** is configured to store a quantity of fuel before, during and/or after operation of the aircraft **12**, including in flight operation and on ground operation. The fuel system **16** is also configured to deliver a flow of the fuel to the aircraft powerplant **14** during operation of the aircraft powerplant **14**. For ease of description, this fuel may be described below as a non-hydrocarbon fuel/a hydrocarbon-free fuel such as hydrogen fuel; e.g., pure H₂. However, it is contemplated the fuel may alternatively be a hydrocarbon fuel, particularly where the hydrocarbon fuel is stored and/or delivered to the aircraft powerplant **14** in a gaseous phase such as natural gas, propane or the like.

(6) The fuel system **16** of FIG. 1 includes a plurality of (e.g., a hydrogen) fuel reservoirs **221-22N**

(generally referred to as “22”) (e.g., fuel storage containers), a (e.g., a hydrogen) fuel control system and a fuel supply circuit 24. The fuel control system of FIG. 1 includes a (e.g., a hydrogen) fuel delivery system 26 and a (e.g., a hydrogen) fuel transfer system 28 which may be discrete from the fuel delivery system 26.

(7) The fuel reservoirs 22 are arranged in an array. The fuel reservoirs 22 of FIG. 1, for example, are fluidly couplable to the fuel supply circuit 24 in parallel through the fuel delivery system 26. Each of the fuel reservoirs 22 is configured to contain and store a respective quantity of the fuel before, during and/or after operation of the aircraft 12. This quantity of the fuel may be stored within the respective fuel reservoir 22 in a liquid phase (e.g., as liquid hydrogen (H.sub.2)) and/or in a gaseous phase (e.g., as hydrogen (H.sub.2) gas). Each fuel reservoir 22, for example, may be configured as or otherwise include a tank, a cylinder, a bladder and/or any other (e.g., insulated) fluid-tight enclosure. Note, while the fuel reservoirs 22 are schematically shown in FIG. 1 in a side-by-side grouping, it is contemplated some or all of the fuel reservoirs 22 may alternatively be arranged at different locations and/or in different compartments of the aircraft 12.

(8) The fuel delivery system 26 is configured to direct a flow of the fuel from the array of fuel reservoirs 22 through the fuel supply circuit 24 to the aircraft powerplant 14. The fuel delivery system 26 of FIG. 1, for example, includes a plurality of reservoir supply circuits 301-30N (generally referred to as “30”), a fuel supply manifold 32 and one or more fuel flow regulators 341-34N (generally referred to as “34”) (“reservoir flow regulators”). Each reservoir supply circuit 30 fluidly couples an outlet from a respective one of the fuel reservoirs 22 to a respective inlet into the fuel supply manifold 32. The fuel supply circuit 24 fluidly couples an outlet from the fuel supply manifold 32 to the aircraft powerplant 14. The fuel supply manifold 32 of FIG. 1 is thereby configured to fluidly couple the reservoir supply circuits 30 and, thus, the fuel reservoirs 22 to the fuel supply circuit 24 in parallel. Each reservoir flow regulator 34 of FIG. 1 is fluidly coupled inline along a respective one of the reservoir supply circuits 30 between the respective fuel reservoir 22 and the fuel supply manifold 32. Each reservoir flow regulator 34 is configured to regulate a flow of the fuel from the respective fuel reservoir 22 to the fuel supply manifold 32 and, thus, the fuel supply circuit 24. Each reservoir flow regulator 34, for example, may selectively fluidly couple or decouple the respective fuel reservoir 22 to or from the fuel supply manifold 32. Each reservoir flow regulator 34 may also or alternatively adjust a flowrate of the fuel from the respective fuel reservoir 22 to the fuel supply manifold 32. Each reservoir flow regulator 34, for example, may be configured as or otherwise include a pump, a control valve or a valve system.

(9) The fuel delivery system 26 may also include a common fuel flow regulator 36 (“supply flow regulator”) for the array of fuel reservoirs 22. This supply flow regulator 36 is fluidly coupled inline along the fuel supply circuit 24 between the fuel supply manifold 32 and the aircraft powerplant 14. The supply flow regulator 36 is configured to regulate a flow of the fuel from the fuel supply manifold 32 to the aircraft powerplant 14. The supply flow regulator 36, for example, may selectively fluidly couple or decouple the fuel supply manifold 32 and, thus, the array of fuel reservoirs 22 to or from the aircraft powerplant 14. The supply flow regulator 36 may also or alternatively adjust a flowrate of the fuel from the fuel supply manifold 32 and, thus, the array of fuel reservoirs 22 to the aircraft powerplant 14. The supply flow regulator 36, for example, may be configured as or otherwise include a pump, a control valve or a valve system. In some embodiments, each reservoir flow regulator 34 may be configured as a control valve and the supply flow regulator 36 may be configured as a pump.

(10) The fuel transfer system 28 is configured to selectively transfer the fuel between some or all of the fuel reservoirs 22. The fuel transfer system 28 of FIG. 1, for example, includes a plurality of reservoir transfer circuits 381-38N (generally referred to as “38”), a fuel transfer manifold 40, one or more transfer valves 421-42N (generally referred to as “42”) and one or more transfer pumps 441-44N (generally referred to as “44”). Each reservoir transfer circuit 38 fluidly couples a transfer aperture of a respective one of the fuel reservoirs 22 to a respective transfer aperture of the fuel

transfer manifold **40**. Note, depending on operation of the fuel transfer system **28**, one transfer aperture may be an outlet and the other transfer aperture may be an inlet. The fuel reservoirs **22** may thereby be fluidly coupled to the fuel transfer manifold **40** in parallel. Each transfer valve **42** of FIG. **1** is fluidly coupled inline along a respective one of the reservoir transfer circuits **38** between the respective fuel reservoir **22** and the fuel transfer manifold **40**. Each transfer valve **42** is configured to regulate a flow of the fuel between the respective fuel reservoir **22** and the fuel transfer manifold **40**. Each transfer valve **42**, for example, may selectively fluidly couple or decouple the respective fuel reservoir **22** to or from the fuel transfer manifold **40**. Each transfer valve **42** may also or alternatively adjust a flowrate of the fuel between the respective fuel reservoir **22** and the fuel transfer manifold **40**. The transfer pumps **44** of FIG. **1** are fluidly coupled inline along the fuel transfer manifold **40**. Each transfer pump **44** of FIG. **1**, for example, is disposed along a portion of the fuel transfer manifold **40** between a neighboring (e.g., adjacent) set of the reservoir transfer circuits **38** and their transfer apertures. These transfer pumps **44** are configured to pump the fuel along the fuel transfer manifold **40** and selectively between any two or more of the reservoir transfer circuits **38**.

(11) The sensor system **18** is configured to identify a fault associated with one or more of the fuel reservoirs **22**. The fault may be indicative of a fuel leak. The fault, for example, may be indicative of a fuel leak from a respective one of the fuel reservoirs **22**; e.g., where the respective fuel reservoir **22** has a crack and/or a loose fitting. In another example, the fault may be indicative of a fuel leak from a fuel line (e.g., one of the circuits **30**, **38**) fluidly coupled to a respective one of the fuel reservoirs **22**; e.g., where the respective fuel line has a crack and/or a loose coupling. Alternatively, the fault may be indicative of an abnormal fuel parameter; e.g., a fuel pressure, a fuel flowrate, etc. The sensor system **18** of FIG. **1** includes a plurality of sensors **461-46N** (generally referred to as “**46**”), where each sensor **46** is associated with (e.g., coupled to, disposed in, located near, etc.) a respective one of the fuel reservoirs **22**.

(12) The controller **20** is arranged in signal communication (e.g., hardwired and/or wirelessly coupled) with the fuel delivery system **26**, the fuel transfer system **28** and the sensor system **18**. The controller **20** may be configured as an onboard engine controller for the aircraft powerplant **14**; e.g., an electronic engine controller (EEC), an electronic control unit (ECU), a full-authority digital engine controller (FADEC), etc. Alternatively, the controller **20** may be arranged in signal communication with the onboard engine controller. The controller **20** may be implemented with a combination of hardware and software. The hardware may include memory **48** and at least one processing device **50**, which processing device **50** may include one or more single-core and/or multi-core processors. The hardware may also or alternatively include analog and/or digital circuitry other than that described above.

(13) The memory **48** is configured to store software (e.g., program instructions) for execution by the processing device **50**, which software execution may control and/or facilitate performance of one or more operations such as those described below. The memory **48** may be a non-transitory computer readable medium. For example, the memory **48** may be configured as or include a volatile memory and/or a nonvolatile memory. Examples of a volatile memory may include a random access memory (RAM) such as a dynamic random access memory (DRAM), a static random access memory (SRAM), a synchronous dynamic random access memory (SDRAM), a video random access memory (VRAM), etc. Examples of a nonvolatile memory may include a read only memory (ROM), an electrically erasable programmable read-only memory (EEPROM), a computer hard drive, etc.

(14) FIG. **2** is a flow diagram of a method **200** of operation for an aircraft system. For ease of description, the operating method **200** of FIG. **2** is described below with respect to the aircraft system **10** of FIG. **1**. The operating method **200** of the present disclosure, however, is not limited to performance with such an exemplary aircraft system.

(15) In step **202**, the aircraft powerplant **14** is operated to facilitate aircraft propulsion (e.g., thrust

and/or lift) and/or power generation. During this powerplant operation, the controller **20** signals the fuel system **16** and its fuel delivery system **26** to deliver a flow of the fuel from the array of fuel reservoirs **22** to the aircraft powerplant **14** through the fuel supply circuit **24**. The fuel delivered to the aircraft powerplant **14** may be provided by (e.g., drawn from) any one or more or all of the fuel reservoirs **22**. The fuel delivery system **26**, for example, may direct fuel from any single one of the fuel reservoirs **22** to the aircraft powerplant **14** until the fuel in that respective fuel reservoir **22** is depleted. The fuel system **16** may then direct fuel from another one of the fuel reservoirs **22** to the aircraft powerplant **14**. Of course, the fuel delivery system **26** may direct fuel from multiple of the fuel reservoirs **22** to the aircraft powerplant **14** when switching between those fuel reservoirs **22** to provide an uninterrupted flow of the fuel to the aircraft powerplant **14**. In another example, the fuel delivery system **26** may direct fuel from multiple of the fuel reservoirs **22** simultaneously.

(16) In step **204**, a fault is identified associated with one of the fuel reservoirs **22**. For example, utilizing sensor data received from one or more of the sensors **46**, the controller **20** may determine presence of a fault associated with, by way of example, the first fuel reservoir **221**. For example, where a pressure within the first fuel reservoir **221** is changing (e.g., decreasing) when that first fuel reservoir **221** is not being utilized or is changing at an unexpected rate (e.g., faster than expected), the controller **20** may identify a fuel leak from the first fuel reservoir **22** (or fuel line coupled thereto). Of course, various other fault identification processes (e.g., fuel leak detections processes) are known in the art, and the present disclosure is not limited to any particular ones thereof.

(17) In step **206**, the faulty fuel reservoir **22** is isolated from the fuel supply circuit **24**. For example, when the fault is identified associated with the first fuel reservoir **221**, the controller **20** may signal the fuel delivery system **26** and its first reservoir flow regulator **341** (e.g., a control valve) to fluidly decouple the first fuel reservoir **221** from the fuel supply manifold **32** and, thus, the fuel supply circuit **24**.

(18) In step **208**, fuel is transferred out of the faulty fuel reservoir **22**. For example, when the fault is identified associated with the first fuel reservoir **221**, the controller **20** may signal the fuel transfer system **28** to transfer at least some or all of the fuel contained within the first fuel reservoir **22** into any one or more or all of the other fuel reservoirs **222-22N**, where this fuel being transferred may be in a liquid phase (e.g., liquid hydrogen (H.sub.2)) and/or a gaseous phase (e.g., hydrogen (H.sub.2) gas). For example, the controller **20** may signal the first transfer valve **421** and the second transfer valve **422** to open. The controller **20** may then signal the first transfer pump **441** to pump some or all of the fuel out of the first fuel reservoir **221** into the second fuel reservoir **222**. In another example, the controller **20** may signal the first transfer valve **421** and the third transfer valve **423** to open. The controller **20** may then signal the first transfer pump **441** and/or the second transfer pump **442** to pump some or all of the fuel out of the first fuel reservoir **221** into the third fuel reservoir **223**. In still another example, the controller **20** may signal the first transfer valve **421**, the second transfer valve **422** and the third transfer valve **423** to open. The controller **20** may then signal the first transfer pump **441** and/or the second transfer pump **442** to pump some or all of the fuel out of the first fuel reservoir **221** into the second fuel reservoir **222** and the third fuel reservoir **223**. Of course, various other variations are possible and contemplated by the present disclosure. Moreover, while the fuel transferring is described above as an active pumping process, it is contemplated the fuel may also or alternatively be passively transferred from the first fuel reservoir **221** to one or more of the other (non-faulty) fuel reservoirs **222-22N** by opening the respective transfer valves **42** to equalize a fuel parameter (e.g., a pressure) of the fuel in the respective fuel reservoirs **22**. By actively or passively transferring the fuel between the fuel reservoirs **22**, the fuel from the first fuel reservoir **221** may be preserved without venting as well as contained in fuel reservoir(s) **222-22N** without a fault.

(19) Before transferring the fuel out of the first fuel reservoir **221**, the controller **20** may determine which of the other (non-faulty) fuel reservoirs **222-22N** is to receive the fuel from the first fuel

reservoir **221**. This selection may be made based on which of the other (non-faulty) fuel reservoirs **222-22N** currently contains the smallest quantity of fuel and, thus, can accommodate some or all of the fuel from the first fuel reservoir **221**. In addition or alternatively, the selection of the fuel reservoir **222-22N** to receive the fuel may be made based on proximity to the first fuel reservoir **221**.

(20) Under certain circumstances, a quantity of the fuel in the first fuel reservoir **221** may be greater than an available capacity of one, some or all of the other (non-faulty) fuel reservoirs **222-22N**. For example, each of the other (non-faulty) fuel reservoirs **222-22N** may already be partially or completely full of fuel. During such circumstances, the controller **20** may temporarily alter (e.g., increase) a predetermined maximum capacity (e.g., pressure capacity) of one or more of the other (non-faulty) fuel reservoirs **222-22N**. The controller **20** may then signal the fuel transfer system **28** to transfer the fuel from the first fuel reservoir **221** into one, some or all of the other (non-faulty) fuel reservoirs **222-22N** until those fuel reservoirs **222-22N** are filled to the new (increased) maximum capacity. In other words, the other (non-faulty) fuel reservoirs **222-22N** may be temporarily over-filled (e.g., over-pressurized) to accommodate the fuel from the first fuel reservoir **221** until the fault associated with the first fuel reservoir **221** can be fixed. In addition or alternatively, the controller **20** may signal the fuel transfer system **28** to vent some of the fuel directed out of the first fuel reservoir **221** into an environment external to the aircraft system **10** and, more generally, external to the aircraft **12** through an optional vent circuit **52**. This venting may be performed concurrently with the transferring of some of the fuel into the other (non-faulty) fuel reservoir(s) **222-22N** or following the transfer of the fuel; e.g., once the other (non-faulty) fuel reservoir(s) **222-22N** are filled to capacity.

(21) During the performance of the steps **206** and **208**, the step **202** may continue to be performed in order to provide a continuous flow of the fuel to the aircraft powerplant **14**.

(22) In some embodiments, an alert may be generated and communicated following the identification of the fault and/or following the transferring of the fuel between the fuel reservoirs **22**. This alert may be provided to personnel operating the aircraft **12** (e.g., a pilot), maintenance personnel, etc. The alert may also be provided to the onboard engine controller such that operation of the aircraft powerplant **14** may be adjusted as needed. Power output from the aircraft powerplant **14**, for example, may be derated to facilitate increased fuel efficiency, etc.

(23) While various embodiments of the present disclosure have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the disclosure. Accordingly, the present disclosure is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. A method of operation, comprising: delivering fuel to a powerplant using a fuel system, the powerplant and the fuel system arranged onboard an aircraft, the fuel system including a first fuel reservoir and a second fuel reservoir, and the first fuel reservoir containing a quantity of the fuel; and transferring at least some of the quantity of the fuel contained in the first fuel reservoir into the second fuel reservoir when a fault is identified associated with the first fuel reservoir; wherein the transferring of the fuel from the first fuel reservoir into the second fuel reservoir increases a second fuel reservoir quantity of the fuel above a predetermined maximum capacity of the second fuel reservoir.

2. The method of claim 1, wherein the fuel comprises hydrogen fuel.

3. The method of claim 1, wherein at least some of the fuel transferred from the first fuel reservoir into the second fuel reservoir is in a gaseous phase.
 4. The method of claim 1, wherein at least some of the fuel transferred from the first fuel reservoir into the second fuel reservoir is in a liquid phase.
 5. The method of claim 1, further comprising identifying the fault associated with the fuel first reservoir.
 6. The method of claim 1, wherein the fault comprises a fuel leak.
 7. The method of claim 1, wherein the fault is indicative of a fuel leak from the first fuel reservoir.
 8. The method of claim 1, wherein the fault is indicative of a fuel leak from a fuel line fluidly coupled to the first fuel reservoir.
 9. The method of claim 1, further comprising: fluidly decoupling the first fuel reservoir from a supply circuit of the fuel system when the fault is identified associated with the first fuel reservoir; and the delivering of the fuel comprising delivering the fuel to the powerplant through the supply circuit.
 10. The method of claim 1, wherein the fuel is delivered to the powerplant using the fuel system before, during and after the transferring of the fuel from the first fuel reservoir into the second fuel reservoir.
 11. The method of claim 1, wherein the transferring comprises pumping at least some of the quantity of the fuel contained in the first fuel reservoir into the second fuel reservoir.
 12. The method of claim 1, wherein the transferring comprises opening a valve between the first fuel reservoir and the second fuel reservoir to equalize a fuel parameter between the first fuel reservoir and the second fuel reservoir.
 13. The method of claim 1, further comprising transferring some of the quantity of the fuel contained in the first fuel reservoir into a third fuel reservoir of the fuel system when the fault is identified associated with the first fuel reservoir.
 14. The method of claim 1, further comprising venting some of the quantity of the fuel contained in the first fuel reservoir to an environment external to the aircraft during and/or following the transferring of the fuel from the first fuel reservoir into the second fuel reservoir.
 15. The method of claim 1, wherein the powerplant comprises a heat engine.
 16. The method of claim 1, wherein the powerplant comprises a fuel cell.
 17. An assembly for an aircraft, comprising: an aircraft powerplant; and a fuel system including a supply circuit, an array of fuel reservoirs and a fuel control system; the array of fuel reservoirs including a first fuel reservoir and a second fuel reservoir, and the first fuel reservoir and the second fuel reservoir fluidly couplable to the supply circuit in parallel; and the fuel control system including a fuel delivery system and a fuel transfer system, the fuel delivery system configured to direct fuel stored in the array of fuel reservoirs to the aircraft powerplant through the supply circuit, the fuel delivery system in fluid communication with the array of fuel reservoirs via outlets of the array of fuel reservoirs, the fuel transfer system configured to transfer at least some of a quantity of the fuel contained in the first fuel reservoir into the second fuel reservoir following identification of a fault associated with the first fuel reservoir while the fuel continues to be directed to the aircraft powerplant through the supply circuit, and the fuel transfer system in fluid communication with the array of fuel reservoirs via transfer apertures of the array of fuel reservoirs; wherein the fuel transfer system is discrete from the fuel delivery system.
 18. The assembly of claim 17, wherein the fuel system is a hydrogen fuel system; the first fuel reservoir is a first liquid hydrogen reservoir; and the second fuel reservoir is a second liquid hydrogen reservoir.
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